

Anne Fayolle

CONTACT INFORMATION	Department of Mathematics University of Utah , LCB 317.	fayolle@math.utah.edu
RESEARCH INTERESTS	Algebraic geometry and commutative algebra.	
EDUCATION	University of Utah PhD in mathematics (2022 - Present) Advisor: Karl Schwede École Polytechnique Fédérale de Lausanne (EPFL) MSc. Mathematics, July 2022 McGill University BSc. First class honours in mathematics, minor in computer science, May 2020 Université de Montréal Certificate Philosophy, with mention of excellence, May 2020	
PUBLICATIONS	2. Anne Fayolle, <i>Centers of perfectoid purity</i> . Journal de l'École Polytechnique Mathématique. DOI:jep.313 2025. 1. J. Carvajal-Rojas, Anne Fayolle, <i>On tame ramification and centers of F-purity</i> . Journal of the London Mathematical Society. DOI: jlms.12993 2024.	
PREPRINTS	1. Anne Fayolle, Abhay Goel, Devlin Mallory, Eamon Quinlan-Gallego, and Teppei Takamatsu. <i>The witt vectors package for Macaulay2</i> . Arxiv Link 2026.	
SOFTWARE	1. The Witt Vector Package for Macaulay2, joint with Abhay Goel, Devlin Mallory, Eamon Quinlan-Gallego, and Teppei Takamatsu.	
SCHOLARSHIPS AND AWARDS	2025 Simons Dissertation Fellowship <i>40 fellowships awarded annually to 3rd year Ph.D students in mathematics by the Simons foundation, \$38,400 over 2 years.</i> 2023 Graduate Student Travel Assistance Award, University of Utah <i>Funds from the department of Mathematics to support travel to a conference.</i> 2023 NSERC CGRS-D, <i>Fellowship awarded by the Canadian government to the top doctoral students applying in the sciences, 105,000 CAD over 3 years.</i> 2022 EDGES fellowship University of Utah <i>Funds awarded by the college of Science at the University of Utah to recruit, retain, and promote the success of outstanding graduate students, \$10,000 over 5 years.</i> 2019 FRQNT supplement to the NSERC award, <i>Fellowship awarded by the Quebec government to undergraduate students to pursue research for a Summer.</i> 2019 NSERC USRA, McGill University, <i>Fellowship awarded by the Canadian government to undergraduate students to pursue research for a Summer.</i> 2017 Alma Mater entrance scholarship, McGill University <i>Merit-based entrance scholarship to first-time university students entering a full-time</i>	

undergraduate degree program.

TEACHING	W 2026	MATH 1210	<i>Calculus 1</i>	Utah
	S 2025	MATH 1210	<i>Calculus 1</i> (Online)	Utah
	W 2025	MATH 1030	<i>Quantitative reasoning</i>	Utah
	W 2024	MATH 1220	<i>Calculus II</i> (Online)	Utah
	W 2023	MATH 1320	<i>Calculus II for engineers</i> (TA)	Utah
SERVICE	2026-2027	AWM (Association for Women in Mathematics) speaker series		
		cochair		
	2025-2026	AWM outreach cochair		
	2024-2025	AWM mentor <i>Mentoring multiple mathematics undergraduates.</i>		
	2024	BRIDGES co-organizer <i>This is an AWM conference aimed towards early graduate students and advanced undergraduates</i>		
	2023-2024	Student commutative algebra seminar organizer		
	2023-2024	Graduate Student Advisory Committee EDI representative		
	2021-2022	Polyquity committee member <i>Polyquity is the association that promotes gender equality on the EPFL campus</i>		
TALKS	2020-2022	EPFL mentor <i>Mentoring a group of first year students to help them navigate the EPFL and math.</i>		
	2019-2020	Vice-President SUMM. <i>This is the largest undergraduate math conference in Canada with over a hundred participants annually.</i>		
	Winter 2027	TBA, Recent developments in commutative algebra, JMM		
	Fall 2026	TBA, University of Michigan, Commutative Algebra seminar		
	Fall 2026	TBA, Arizona State University AMS Sectional		
	Summer 2026	Perfectoid pure pairs, Université Paris Saclay, Informal Seminar		
	Winter 2026	<i>F-singularities of foliations</i> , North Dakota State University AMS Sectional		
	Winter 2026	<i>Centers of perfectoid purity</i> , Special session on commutative algebra, JMM		
	Fall 2025	<i>Centros de pureza perfectoides</i> (in Spanish), Seminario Chichimeca de Álgebra, UA-Aguascalientes		
	Fall 2025	<i>Centers of perfectoid purity</i> , Algebra seminar, University of Missouri		
	Fall 2025	<i>Centers of perfectoid purity</i> , Workshop on Notions of Singularity in Different Characteristics, Banff International Research Station		
	Fall 2025	<i>Centers of perfectoid purity</i> , Fall Southeastern AMS sectional, Tulane University		
	Fall 2024	<i>Centers of perfectoid purity</i> , CIMAT Algebra seminar		
	Fall 2024	<i>Centers of perfectoid purity</i> , Congreso nacional de la sociedad matemática mexicana		
	Summer 2024	<i>Mixed characteristic compatible ideals</i> , Upcoming Researchers in Commutative Algebra, University of Nebraska Lincoln		
	Spring 2024	<i>Tame ramification and centers of F-purity</i> , CIMAT Commutative Algebra Learning Seminar, CIMAT		
	Fall 2023	<i>Tame ramification and centers of F-purity</i> , Commutative Algebra Seminar, University of Utah		
	Fall 2023	<i>On tame ramification and compatible ideals</i> , BIKES, the Graduate Student Seminar at the University of Utah		
	Fall 2023	<i>Compatible ideals and their behavior under finite covers</i> , Commutative Algebra Reading Seminar, University of Nebraska		

Spring 2023 *An introduction to compatible ideals*, Commutative Algebra Graduate Student Seminar at U Michigan
Spring 2023 *An introduction to compatible ideals*, BIKES, the Graduate Student Seminar at the University of Utah

POSTERS *Centers of perfectoid purity*, KUMUNU, University of Missouri, Fall 2025
Centers of perfectoid purity, SRI Bootcamp, University of Colorado Fort Collins, Summer 2025
Centers of perfectoid purity, Singularities in Algebra and Geometry, CIMAT, Summer 2025
Tame ramification and centers of F -purity, Western Algebraic Geometry Symposium, Washington University in St. Louis, Fall 2023.